

Comprehensive Final Audit

What is prohibited, what is missing, where each gap must be fixed, and what remains before an independently built N-PORT filing can be released.

As of August 3, 2026 · Source boundary corrected

Controlling rule: no file, value, mapping, or pre-fill delivered by U.S. Bank may populate the in-house filing. Those artifacts are read-only comparison evidence.

1. Final conclusion

No June 2026 output is release-eligible. The repository's landed custody, EagleSTAR, prepared N-PORT, master workbooks, and derived per-fund files came from U.S. Bank-delivered material. They may be compared against independently generated output, but they cannot populate it.

The correct comparison is output-to-output after dates and fund identity are aligned. It is not a source merge. Numerical differences caused only by different report dates are not gaps. A gap is missing coverage, evidence, logic, or reconciliation in the in-house service.

2. Missing versus blocking

Term	Meaning	Example	Release effect
Missing	A required value, dataset, approval, or proof is absent.	No internal position close exists for 2026-06-30.	Usually becomes a blocker if applicable.
Blocking	The system must refuse release. The cause may be missing, stale, invalid, unapproved, prohibited, or unreconciled.	A netAssets value is present but came from U.S. Bank.	Release must stop.
Non-blocking review	Optional or supported data needs confirmation, but omission is permitted by the filing contract.	ISIN is absent but a valid CUSIP is filed.	Review and document; does not automatically stop release.

Key point: “present” does not mean “usable.” A value from a prohibited source is a blocker even when it is populated.

3. What the code does now

Control	Current behavior	Truth / limitation
Production build	<code>nport build</code> reads canonical <code>fund_config.txt</code> , <code>filing_data.txt</code> , and <code>holdings.csv</code> only.	It no longer calls the U.S. Bank custodian or EagleSTAR parsers.
Fund review	<code>prepare-review</code> creates one workbook under the fund and period.	It contains exact locations for every fund field and surfaced holding exception.
Evidence gate	<code>review-status</code> checks values, dispositions, sources, cutoffs, hashes, counts, timestamps, and approvals.	Known U.S. Bank/EagleSTAR labels and legacy repository paths are rejected.
Clean finalization	<code>finalize-review</code> writes a new versioned bundle under <code>data/</code> builds.	It never overwrites the source workbook or legacy canonical files.
Legacy/write workflow	<code>masters</code> , <code>split</code> , <code>mergehumanreview</code> , <code>enrich</code> , <code>merge</code> , <code>new-filing</code> , and the master-specific commands exit with an error.	They either consume prohibited material or cannot authenticate independent origin.
Reference comparison	<code>nport compare-reference</code> is read-only.	It reports structural differences and never writes filing inputs.
Residual limitation	Manifest checks use labels, paths, dates, hashes, counts, and approval.	They cannot prove origin if a prohibited file is deliberately moved and relabelled; authenticated connectors remain required.

4. Complete open-blocker register

These are the 13 tracked gaps. Engineering status and data status are separate: a workflow can be implemented while the filing remains blocked because the independent value has not been supplied.

File-name rule: the names below are expected internal naming templates, not claims that the files currently exist. Use the real source filename when different. An empty placeholder never closes a blocker.

ID / status	Gap / affected fields	Required file templates
G-001 DATA BLOCKER	Independent position population All Part C holdings, including GPTZ if the fund owns it	<code>data/intake/<period>/<fund>/positions.csv</code> clean bundle: <code>holdings.csv</code>
G-002 DATA BLOCKER	Independent fund-accounting close Assets, liabilities, net assets, balance-sheet items, gains, returns, and cash	<code>accounting_close.csv</code> or <code>accounting_close.pdf</code> clean bundle: <code>filing_data.txt</code>

G-003 PARTLY AUTOMATED	Create/redeem flows mon1/2/3 Sales, Redemption, and Reinvestment	create_redeem_orders.csv reinvestment_support.csv (if applicable)
G-004 ENGINEERED	Fund policy and static data Fiscal year-end, filing type, derivatives regime, applicability, identity, and signer fields	fund_policy_approval.pdf fund_static_data.csv clean bundle: fund_config.txt
G-005 ENGINEERED	Source manifest Lineage for every accepted source	human_review.xlsx clean bundle: source_manifest.csv
G-006 ENGINEERED CONTROL	Prohibited comparison inputs U.S. Bank, EagleSTAR, prepared filings, legacy masters, and derived canonical files	comparison only: data/custodian/* data/fund_accounting/* data/master/*.xlsx legacy data/funds/* files
G-007 ENGINEERED	Human-review evidence Value/disposition, source, cutoff, reviewer, time, reason, and two-person approval	human_review.xlsx field_provenance.csv review_receipt.json
G-008 DATA BLOCKER IF SWAPS	Swap economics Counterparty, LEI, notional, value, termination, and receive/pay leg terms	swap_trade_export.csv executed_swap_confirmations.pdf
G-009 DATA BLOCKER IF APPLICABLE	B.2.f / B.3 / B.5.c / B.9 / B.10 Cash outside C/D, risk metrics, monthly categories, derivatives exposure, or VaR	cash_ledger.csv risk_metrics.csv rule_18f4_results.csv var_backtesting.csv monthly_return_categories.csv
G-010 DATA BLOCKER IF APPLICABLE	C.7 liquidity liquidityClassificationJson and liquidityCircumstancesJson by holding	liquidity_classifications.csv liquidity_circumstances.csv (if applicable)
G-011 DATA BLOCKER FOR OPTIONS	Option terms and delta Delta, exercise terms, expiry, and underlying/reference identity	option_trade_terms.csv option_delta.csv
G-012 PARTLY ENGINEERED	Reconciliation NAV equation and position count are automated; GL, derivatives, and flow tie-outs still need source reports	reconciliation.csv positions_to_gl_reconciliation.csv flow_reconciliation.csv derivatives_reconciliation.csv
G-013 RESIDUAL CONTROL LIMIT	Origin assurance Local files cannot prove a reviewer did not manually copy and relabel a prohibited value	source_origin_attestation.pdf review_receipt.json

Exact fix location and proof

ID	Exactly where to fix	Owner	Proof to close
G-001	prepare-review <fund> <period> --positions <independent.csv>; review surfaced rows in HoldingFields.	Operations	Approved POSITIONS row, period-end cutoff, hash, row count, and position-count reconciliation
G-002	Add the internal close file to Sources; enter supported values in FundFields where targetFile=filing_data.txt.	Fund Accounting	Each row cites the approved source; NAV equation passes
G-003	Pass --orders to prepare-review. Review calculated sales/redemptions in FundFields. Enter reinvestment only from separate internal evidence.	Capital Markets + Accounting	Accepted-order calculation, cutoff review, and independent reconciliation

G-004	Use FundFields where targetFile=fund_config.txt. No separate fund_registry.csv is required.	Compliance / Legal	Approved value, effective date, approver, timestamp, and internal policy reference
G-005	Complete Sources. Finalize-review creates source_manifest.csv automatically.	Operations	Path exists; cutoff, hash, count, preparer, reviewer, and approval are complete
G-006	Do not add them to Sources. Known labels and repository landing paths are rejected.	Engineering + Operations	Prohibited-source check passes; legacy writers remain disabled
G-007	Complete FundFields, HoldingFields, Sources, and Approvals in the fund-period workbook.	Operations	review-status returns zero blockers; review receipt hashes workbook and outputs
G-008	Use HoldingFields for the swap record. Cite internal trade evidence or an independently retained executed confirmation.	Derivatives Operations	All required swap fields supported; valuation reconciles
G-009	Use FundFields. Enter a supported value or NOT_APPLICABLE with source, reviewer, time, and reason.	Accounting + Risk	Fund policy determines applicability; preflight enforces applicable sections
G-010	Use HoldingFields for each surfaced holding and cite internal liquidity-risk evidence.	Liquidity Risk	100% applicable holding coverage and approval evidence
G-011	Use HoldingFields for the option record and cite internal trade/risk evidence at the cutoff.	Derivatives Risk	Required option fields supported and validated
G-012	Clear automated results in Reconciliation and attach the other internal control reports in Sources.	Accounting + Operations	No automated exception; independent control reports reviewed
G-013	Use source-system attestations, hashes, dual review, and the receipt. This remains a governance control without authenticated connectors.	Operations + Compliance	Documented origin attestation; no claim that local code can prove origin

5. Where a human enters values

Implemented location: `data/funds/<fund>/filings/<period>/human_review.xlsx`. This is the only workbook used by the new review path.

Sheet	Row key	Editable values	Covers	Release rule
Summary	Fund + period	Nothing	Scope, stop rule, and next command	Read-only
GapGuide	gapId	Read-only explanation	All 13 gaps and exact sheet	Navigation only
FundFields	targetFile + fieldName	proposedValue, sourceId, sourceAsOf, status, reviewer, reviewedAt, comment	Fund config and filing-level values	APPROVED value or supported NOT_APPLICABLE disposition

HoldingFields	recordKey + fieldName	Same review columns	Debt, derivative, option, swap, and liquidity exceptions	Generated from the independent positions file
Sources	sourceId	System, path, cutoff, hash, count, preparer, reviewer, approval	Evidence used by reviewed rows	U.S. Bank/reference sources are rejected
Approvals	PREPARER / REVIEWER	Name, approvedAt, status, comment	Package approval	Two different people required
Reconciliation	check	Status, actual, expected, detail	NAV equation and position count	Generated during evaluation/finalization

The workbook is generated for one fund and one period from the independently supplied positions file. It is a controlled queue: Sources establishes the evidence, FundFields and HoldingFields record decisions, Approvals records two-person sign-off, and Reconciliation records automated results. It is not a source of facts by itself.

What a human may and may not do

- May: enter a value supported by an approved independent source.
- May: mark a conditional field NOT_APPLICABLE only with a source, reviewer, timestamp, and reason.
- May not: invent a zero, N/A, policy choice, identifier, term, or classification.
- May not: cite or copy U.S. Bank, EagleSTAR, a prepared filing, or a file derived from them.
- May not: type the base position population into Excel; it must be supplied with `--positions`.

6. Correct operating sequence

Step	Actor	Deterministic action	Completion condition
1. Intake	Operations	Place the actual independent positions file and optional orders file in the fund-period intake folder.	Real files exist; their origin is independent of U.S. Bank.
2. Template	Operations	<code>nport prepare-review <fund> <period> --positions <file> [--orders <file>]</code>	The fund-period <code>human_review.xlsx</code> exists.
3. Evidence	Preparer + source owner	Complete Sources with the actual system, path, cutoff, hash/count, and preparation evidence.	Every used source row is complete and PENDING or APPROVED.
4. Decisions	Field owners	Resolve generated FundFields and HoldingFields rows; every accepted decision cites Sources.	No applicable row is MISSING or NEEDS_REVIEW.
5. Approval	Second reviewer	Review sources and decisions; complete Approvals separately from the preparer.	PREPARER and REVIEWER are different and both APPROVED.
6. Gate	Operations	<code>nport review-status <fund> <period></code>	Command reports zero blockers.
7. Build	Operations	<code>nport build <fund> <period> --from-review</code>	Versioned clean bundle and validated XML are written.
8. Compare	Reviewer	Compare the independent XML with U.S. Bank output on the same fund/date only.	Differences are documented; comparison data is never merged back.

All-fund operation: repeat the sequence independently for every fund. The runbook contains copy/paste

PowerShell loops that discover fund folders from `data/intake/<period>/`, create one workbook per fund, stop on blocked reviews, and build only review-ready funds. FDRS is an example ticker, not a scope limitation.

Field coverage: the runbook contains a generated trace for all 170 accepted canonical fields: 30 fund-configuration fields, 56 filing-level fields, and 84 holding fields. It separates the actual current/legacy writer from the new review location.

7. What remains before release

The local review, evidence, merge, provenance, basic reconciliation, and build controls are implemented. What remains is the actual independent fund data and control evidence for each period, plus GL/position, flow, and derivatives reconciliations. Local code also cannot prove that a person did not copy and relabel a prohibited value. Until the applicable data and evidence are supplied, the honest filing status remains **BLOCKED**.

Generated from the repository field contracts and source controls on 2026-08-03.